Prakash Laxman Suryawanshi
Founder & Director, Qodeigence Infocircle Private Limited
Maharashtra, India
B.E. Computer Engineering, Sinhgad Institute of Technology, Lonavala
prakashsurywanshi1@live.com

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The Editor-in-Chief
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Subject: Submission of an original research manuscript — "A Sustainable Hybrid QKD–PQC Security Architecture for Quantum-Resilient Residential IoT"

Dear Editor,

I am pleased to submit the original single-author research manuscript entitled "A Sustainable Hybrid QKD–PQC Security Architecture for Quantum-Resilient Residential IoT" for consideration in the *IEEE Internet of Things Journal*. The manuscript has not been published previously and is not under consideration for publication elsewhere.

Scope and Significance

Battery-powered residential IoT nodes must remain operational for years while facing the emerging threat of quantum cryptanalysis and the "harvest-now, decrypt-later" attack model. Migrating conventional public-key cryptography to standardized post-quantum cryptography (PQC) directly on constrained leaf nodes, however, imposes a severe energy and radio-bandwidth penalty because large lattice-based keys are split across many IEEE 802.15.4/6LoWPAN frames. This paper addresses that sustainability tension with a gateway-centric hybrid architecture that keeps asymmetric post-quantum operations off the leaf nodes while preserving long-term post-quantum resilience. The work fits the Journal's scope at the intersection of IoT security, edge/energy sustainability, and next-generation network protocols.

Key Points and Contributions

1. **Sustainable hybrid QKD–PQC architecture.** An edge-centric Quantum-Safe Home Security Gateway (QHSG) structurally isolates security-intensive key-management operations from high-volume application data streams; last-mile Quantum Key Distribution (QKD) material is ingested where optical infrastructure exists, while standardized ML-KEM-768 (FIPS 203) provides a deployable path over conventional networks. Bulk data is protected by authenticated symmetric encryption.
2. **Grounded security–sustainability model.** Endpoint compute energy, radio transmission overhead, and 6LoWPAN fragmentation penalties are formalized as explicit design param-

eters. Constants are anchored to the IEEE 802.15.4 physical layer and the TI CC2652R datasheet (P_{TX}=22 mW, P_{RX}=21 mW at 250 kbps), with compute times from measured ARM Cortex-M4 cycle counts.

3. **Quantified energy outcome.** Direct leaf-node PQC deployment raises endpoint key-negotiation energy by +108.4% over classical handshakes (328.46 mJ vs. 157.63 mJ at N=100) because the 2272-byte ML-KEM-768 handshake fragments into 24 link-layer frames. The proposed gateway-mediated hybrid reduces endpoint key-management energy by −96.8% relative to standalone PQC and −93.3% relative to classical ECDH.

4. **Protocol-level validation.** The fragmentation model is validated in *executed* headless Contiki-NG/COOJA runs across seven payload sizes (64–2272 B); the measured frame counts (1, 2, 3, 6, 11, 13, 24) match the closed form $\lceil(\text{payload} + 12)/96\rceil$ exactly.

5. **Formal security verification.** A suite of ProVerif 2.05 models machine-checks key-lifecycle secrecy and authentication, single-use refresh semantics, endpoint isolation under device compromise, gateway-compromise forward secrecy, harvest-now-decrypt-later resistance, cross-session/device key independence, ticket replay under packet loss, and active local-link impersonation.

6. **Reproducibility.** Every quantitative and symbolic result is regenerable from a public, tagged artifact containing one run script per evidence stream, including the Contiki-NG patch required to transmit the 2272-byte handshake.

Data and Reproducibility

All simulation code, grounded parameter sets, figure-generation routines, sensitivity-analysis scripts, ProVerif models, cycle-calibration scripts, and Contiki-NG/COOJA scenarios are openly archived at:

https://github.com/prakashsurywanshi/IEEE_quantum_safe_residential_iot
(tag `v1.0.0`)

so that reviewers can independently reproduce each claim.

I believe this work offers a rigorous, hardware-grounded, and formally verified framework for sustainable quantum-resilient IoT security, and I hope it is found suitable for publication in the *IEEE Internet of Things Journal*. I would be glad to provide any additional information or artifacts requested.

Yours sincerely,

Prakash Laxman Suryawanshi
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